

Banking & Financial Performance Analytics

Synthetic portfolio | 2025 | INR

Metric	Result
Customers	2,500
Deposit balance (31 Dec)	INR 52.32 crore
Loan outstanding (31 Dec)	INR 89.98 crore
Loan-to-deposit ratio	172.0%
90+ DPD exposure ratio (proxy)	36.0%
Annual operating profit before losses/tax	INR 6.31 crore
Clean annual transactions	40,000

Findings

Chennai Central has the largest deposit balance: INR 7.02 crore, 13.4% of the portfolio.

Deposits changed -0.5% from the 1 January opening balance to 31 December.

Mobile and Internet account for 71.7% of annual transactions.

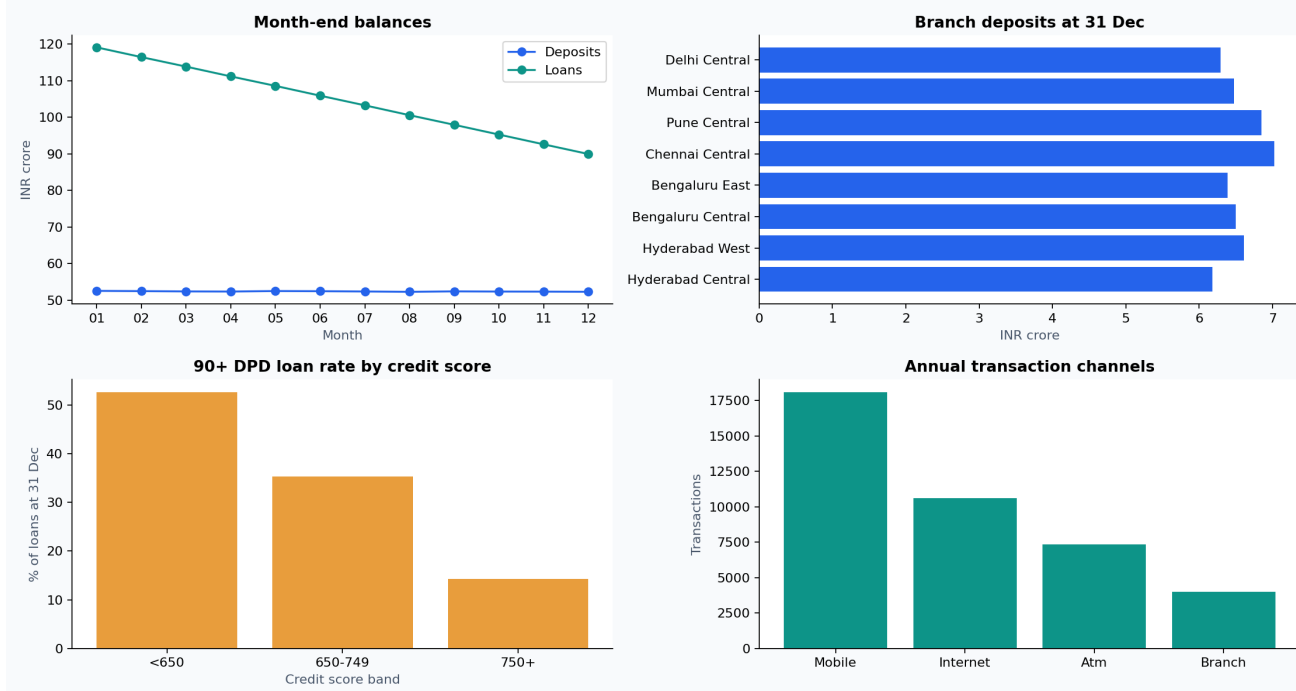
The <650 score band has the highest 90+ DPD loan rate (52.6%, 266 loans). This pattern is partly programmed into the simulation.

90+ DPD balances represent 36.0% of outstanding loans. This is an analytical proxy, not a regulatory NPA classification.

Annual management operating profit before credit losses and tax is INR 6.31 crore.

Portfolio dashboard

BANKING PERFORMANCE | Synthetic 2025 portfolio



Recommended next steps

Prioritize a collections review of 90+ DPD loans, recording recoveries and arrears before interpreting risk trends.

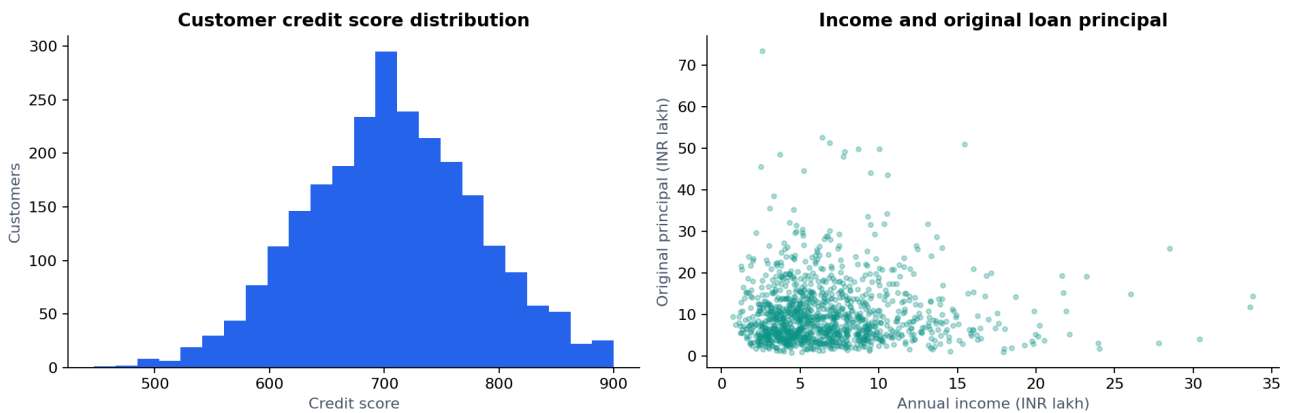
Review funding needs alongside the loan-to-deposit ratio. Wholesale funding expense is explicitly modeled for branches with a funding gap.

Compare branch operating costs and income before reallocating resources. Test any proposed change against real operational constraints.

Replace simulated patterns with appropriately anonymized real data before making a banking decision.

Methodology and validation

Data quality: raw transactions: 40115, exact duplicates removed: 100, transactions quarantined: 15, clean transactions: 40000, channel aliases normalized: 40, missing occupations labelled unknown: 20



All customers, transactions and results are generated with seed 81

The loan schedule uses equal principal installments, not fixed-EMI amortization

Missed installments remain unpaid through year-end; later scheduled payments do not clear older arrears

Interest income is cash received

Deposit cost uses average opening/closing monthly balances and a 3.5% annual rate

Wholesale funding uses an 8% annual rate for positive branch loan-minus-deposit gaps

Fees are separately simulated at INR 18 per transaction

Operating profit excludes credit-loss provisions and tax

Fee and deposit-interest postings are outside the customer transaction ledger

No financial prediction or regulatory compliance is claimed.

All nine pipeline validation checks passed: transaction accounting, account and loan roll-forwards, branch deposit and profit reconciliation, non-negative balances, foreign keys and DPD bounds.